

- HII: (i) ionization  $\leadsto S_{\star}$   
(ii) recombination  $\leadsto R_{\text{tot}}$   
(iii)  $R_s$

• last:  $R_{\text{tot}}$ : assuming constant  $n_H$

• PS:  $n_H \sim R^{-3}$

P11:  $\frac{I_1}{I_2} = (100)^{\frac{1}{5}(m_2 - m_1)}$  \*

$\nearrow I$  increases

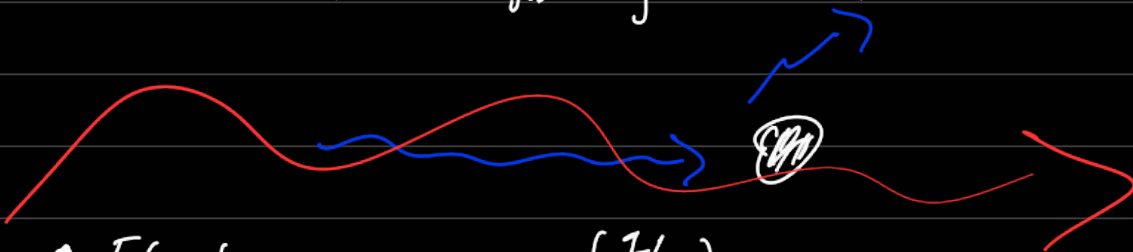
magn.

•  $I_1 = L_1 / 4\pi D_1^2 \leftarrow D_1$ : Distance to object 1

\*  $A_\lambda$ :

PS11 • lower  $\lambda$ : get scattered

• Blue light get scattered



• Electrons: spin  $[1/2]$

light: polarization spin 1

• Once light has left the source:

Dust will become ionized by high energy photons

$$\langle E \rangle = h\nu \sim kT$$

(i) Some of the light will absorbed in its travel to Earth.

\* From this we get HII regions (regions of ionized hydrogen  $H^+ + e^-$ )

$\rightarrow$  light will interact with more frequently with the ions and the electromagnetic fields in this region

This will cause

- (ii) More light to scatter by there being more particles  
 (iii) This scattering, will reddening the light since blue light is comparable in size to the dust. This is measured by the extinction

$A_\lambda$

$$m - M = 5 \log(D) - 5 + A_\lambda$$

- In our notes

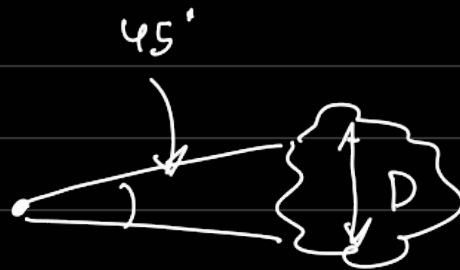
$$A_\lambda \sim \lambda^{-1} = \frac{1}{\lambda} (\text{Dust})$$

$$A_\lambda \sim \lambda^{-4} = \frac{1}{\lambda^4} (\text{our atmosphere})$$

- (iv) As we observe the light: our device measures a polarization of light. From this we can deduce that the light's polarization is affected by the dust.

→ Comes from EEM interactions within the ionized region

PS1 | optimal  
 (d) HII



$$\rightarrow 45' = \theta \quad \theta'' = \frac{1}{D \text{ pc}}$$

$$(45 \text{ arcmin}) \left( \frac{60 \text{ arcsec}}{1 \text{ arcmin}} \right) = 45 \cdot 60''$$

$$D \text{ pc} = \frac{1}{45 \cdot 60}$$

(e) H<sub>B</sub>:

Even easier :  $I_{\text{obs}} = 3.5 \times 10^{-13} \text{ W m}^{-2}$

$$m_{\text{observed}} = m_{\text{true}} + A_\lambda$$

$$m_{\text{obs}} - m_{\text{true}} = A_{\lambda} \quad \swarrow = A_{\lambda}$$

Since

$$\frac{I_{\text{true}}}{I_{\text{obs}}} = 100^{\frac{1}{5} (m_{\text{obs}} - m_{\text{true}})}$$

$$\log_{10} \left( \frac{I_{\text{true}}}{I_{\text{obs}}} \right) = \frac{2}{5} A_{\lambda}$$

- $I_{\text{obs}}$  is given to us
- $I_{\text{true}}$  we find from (c) and (d)

$$I_{\text{true}} = \frac{L}{4\pi D^2}$$

Found in (c)  
Found in (d)

$$\frac{5}{2} \log_{10} \left( \frac{\frac{L}{4\pi D^2}}{I_{\text{obs}}} \right) = A_{\lambda}$$

$$\boxed{2.5 \log_{10} \left( \frac{L}{4\pi D^2 I_{\text{obs}}} \right) = A_{\lambda}}$$

